

Colorings with Fractional Defect

Wayne Goddard and Honghai Xu

Dept of Mathematical Sciences, Clemson University

Abstract

Consider a coloring of a graph such that each vertex is assigned a fraction of each color, with the total amount of colors at each vertex summing to 1. We define the fractional defect of a vertex v to be the sum of the overlaps with each neighbor of v , and the fractional defect of the graph to be the maximum of the defects over all vertices. Note that this coincides with the usual definition of defect if every vertex is monochromatic. We provide results on the minimum fractional defect of 2-colorings of some graphs.

1 Introduction

In a usual vertex coloring of a graph, every vertex is assigned one color and that color is different from each of its neighbors. We consider here a two-fold generalization of this: a vertex can receive multiple colors and can overlap slightly with each neighbor. Specifically, each vertex is assigned a fraction of each color, with the total amount of colors at each vertex summing to 1. The **(fractional) defect** of a vertex v is defined to be the sum of the overlaps over all colors and all neighbors of v ; the **(fractional) defect** of the graph is the maximum of the defects over all vertices. Note that if every vertex is monochromatic (has only one color), then our fractional defect coincides with the usual definition of defect (see for example [2]); and that defective colorings are also called improper colorings.

The idea of assigning vertices multiple colors has been used most notably in fractional colorings (e.g. [8, 5]), but also for example in t -tone colorings [4]. Like in t -tone colorings (and unlike in fractional colorings), we consider here the situation where one pays for each color used, regardless of how much the color is used. Note that for proper colorings, allowing one to color a vertex with multiple colors does not yield anything new. For, one can just choose for each vertex v one color present at v and recolor it entirely that

⁰Corresponding author: Wayne Goddard, goddard@clemson.edu, phone +1-864-656-0186, fax +1-864-656-0145

color, and therefore the minimum number of colors needed is just the chromatic number. Similarly, with the usual definition of the defect of a vertex as the number of neighbors that share a color, there is no advantage to using more than one color at a vertex. But we consider colorings where a vertex overlaps only slightly with each neighbor.

Consider, for example, the Hajós graph. Figure 1 gives a 2-coloring of this graph with defect $4/3$ (and this is best possible in that any 2-coloring has at least this much defect). For another example, consider the complete graph on 3 vertices. Any 2-coloring of K_3 has defect at least 1, but there are multiple optimal colorings: color one vertex red, one vertex blue, and the third vertex any combination of red and blue.

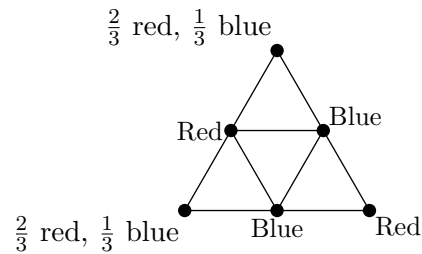


Figure 1: An optimal 2-coloring of the Hajós graph

Our objective is to minimize the defect of the graph. Specifically, for a given number of colors, what is the minimum defect that can be obtained? If the number of colors is the chromatic number, then of course there need be no defect. But if the number of colors is smaller, then there is a defect.

In the rest of the paper we proceed as follows. In Section 2 we introduce some notation and observe some elementary results about monochromatic vertices. Thereafter in Section 3 we consider calculating the optimal 2-colorings for several graph families, including fans, wheels, complete multipartite graphs, and rooks graphs. Thereafter we observe that the decision problem is NP-hard, and pose some more conjectures.

2 Preliminaries

Consider a graph G and k a number of colors. For color j , let $f_j(v)$ be the usage of color j on vertex v . Then the **defect** of vertex v is given by

$$d(v) = \sum_{w \in N(v)} \sum_{j=1}^k \min(f_j(v), f_j(w)). \quad (1)$$

In general, the problem is to minimize

$$\max_v d(v)$$

over all colorings such that $f_j(v)$ is nonnegative and $\sum_{j=1}^k f_j(v) = 1$ for all vertices v . We denote this minimum by $D(G, k)$, and call it the **minimum defect**.

Note that the existence of the minimum is guaranteed, since the objective function is continuous and the feasible region is a closed bounded set. Further, the calculation is at least finite, since, for example, we can prescribe which of $f_j(v)$ and $f_j(w)$ are smaller in every min in Equation 1 for each vertex v , and thus $D(G, k)$ is the minimum over exponentially many linear programs.

A related parameter is the **minimum total defect**: we define $TD(G, k)$ to be the minimum over all colorings of $\sum_{v \in V} d(v)$. However, here fractional colorings do not yield anything new:

Lemma 1 *For graph G and number of colors k , there is a k -coloring that achieves $TD(G, k)$ in which every vertex is monochromatic.*

Proof. Consider any vertex v that is not monochromatic: say $f_1(v), f_2(v) > 0$ with $f_1(v) + f_2(v) = A$. Then consider adjusting the coloring such that $f_1(v) = x$ and $f_2(v) = A - x$. As a function of x , the defect of v with any neighbor w is a (piecewise-linear) concave-down function. Thus, the total defect of the graph, as a function of x , is a concave-down function, and so its minimum is attained at an endpoint. This means that one can either replace color 1 by color 2 or replace color 2 by color 1 at v without increasing the total defect. Repeated application of this replacement yields a coloring with every vertex monochromatic. $\square$

There are several fundamental results about monochromatic vertices for minimum defect. One is that we may assume that there is a monochromatic vertex of each color.

Lemma 2 *Let k be an integer and G be a graph with at least k vertices. Then there is a k -coloring that achieves $D(G, k)$ that has at least one monochromatic vertex for each color.*

Proof. Consider any optimal k -coloring, and consider each color $j = 1, 2, \dots, k$ in turn. Each time, define vertex v_j as any vertex other than $v_1, \dots, v_{j-1}$ with the largest usage of color j ; then recolor v_j (if needed) such that $f_j(v_j) = 1$ and $f_i(v_j) = 0$ for all $i \neq j$. Such a recoloring does not increase the defect at any vertex. So we will reach an optimal coloring with the desired property. $\square$

We next show that the minimum defect is either 0 or at least 1.

Lemma 3 *For any graph G and positive integer k , if $D(G, k) > 0$ then $D(G, k) \geq 1$.*

Proof. If every vertex is monochromatic, then the defect is an integer and so the result is immediate. So consider any vertex v that is not monochromatic. If for any color j we have $f_j(v) \geq f_j(w)$ for all neighbors w of v , then we can recolor v to be monochromatically color j without increasing the defect of any vertex. So we may assume that for every color j at v , vertex v has a neighbor w_j with $f_j(w_j) \geq f_j(v)$. It follows that

$$d(v) = \sum_{w \in N(v)} \sum_{j=1}^k \min(f_j(v), f_j(w)) \geq \sum_{j=1}^k \min(f_j(v), f_j(w_j)) = \sum_{j=1}^k f_j(v) = 1.$$

The result follows. $\square$

For example, it follows from Lemma 3 that the optimal 2-coloring of the odd cycle C_n has defect 1.

Proposition 1 *The complete graph K_n has $D(K_n, k) = \lceil n/k \rceil - 1$, and this is best possible.*

Proof. This defect is achieved by (inter alia) coloring each vertex with a single color and using each color as equitably as possible. (This is trivially the best coloring for total defect.)

To see that $\lceil n/k \rceil - 1$ is best possible, we proceed by induction on n , noting that the result is trivial if $n \leq k$. So assume $n > k$. By Lemma 2, there is an optimal k -coloring

with for each j a vertex v_j that is monochromatically color j . Let $A = \{v_1, \dots, v_k\}$. Then the defect of any other vertex w in G equals 1 plus the defect of w in $G - A$. By the induction hypothesis, there exists a vertex in $G - A$ that has defect at least $\lceil (n - k)/k \rceil - 1$ in $G - A$. This proves the lower bound. $\square$

3 Two-Colorings of Some Graph Families

We now consider results for 2-colorings. Unless other specified, we assume the colors are red and blue, and denote the red usage at vertex v by $r(v)$ (so that the blue usage is $1 - r(v)$).

3.1 Fans

The *fan*, denoted by F_n , is the graph obtained by joining one vertex to a path of order n .

Lemma 4 *In any 2-coloring of F_3 it holds that $d(v) + d(w) \geq 2$ where v and w are the dominating vertices.*

Proof. Suppose the dominating vertices are v and w and the other two vertices are a and b . Let e_{xy} denote the overlap between vertices x and y . Then $d(v) + d(w) = e_{va} + e_{vb} + e_{wa} + e_{wb} + 2e_{vw}$; further, because a triangle has total defect at least 2, we have $e_{va} + e_{vb} + e_{vw} \geq 1$ and $e_{wa} + e_{wb} + e_{vw} \geq 1$. The result follows. $\square$

Note that F_1 is just K_2 and F_2 is just K_3 , and so it holds that $D(F_1, 2) = 0$ and $D(F_2, 2) = 1$. For the general cases of F_n , we have the following:

Proposition 2 *The defect of F_n ($n \geq 3$) is*

$$D(F_n, 2) = \frac{2\lfloor n/3 \rfloor}{\lfloor n/3 \rfloor + 1}.$$

Proof. Say the path is $v_1 v_2 \dots v_n$ with dominating vertex h .

We prove the upper bound by the following construction. Set $x = 2/(\lfloor n/3 \rfloor + 1)$. Let $r(h) = 1$. Let $r(v_i) = x$ if i is a multiple of 3, and 0 otherwise. It can readily

be checked that every vertex v_i has defect at most $2 - x$, and that vertex h has defect $\lfloor n/3 \rfloor x$. The result follows since these two values are equal.

To prove the lower bound, it suffices to show that $D(F_n, 2) \geq 2n/(n+3)$ if n is a multiple of 3. We partition the path P_n into $n/3$ copies of P_3 ; thus each P_3 along with vertex h forms a copy of F_3 . It follows from Lemma 4 that $d(w) + \sum_{i=1}^{n/3} d(v_{3i-1}) \geq 2n/3$, whence the result. $\square$

Note that the defect $D(F_n, 2)$ tends to 2 as n increases. The fan is outerplanar. Several researchers [1, 7] showed that one can ordinarily 2-color an outerplanar graph with defect at most 2. However, we conjecture that this bound can be improved slightly, in that for any outerplanar graph G it holds that $D(G, 2) < 2$.

3.2 Wheels

The **wheel**, denoted by W_n , is the graph formed by joining a single vertex to every vertex of a cycle of order n . The vertex of degree n is called the **hub** of the wheel.

We have the following construction.

Proposition 3 *The wheel W_n can be 2-colored with defect*

$$D(W_n, 2) \leq \frac{2\lceil n/3 \rceil}{\lceil n/3 \rceil + 1}.$$

Proof. Let $x = 2/(\lceil n/3 \rceil + 1)$ and let D be a minimum independent dominating set of the cycle. For a vertex v on the cycle, let $r(v) = x$ if $v \in D$, and $r(v) = 0$ otherwise. Let $r(h) = 1$ for the hub h . It can readily be checked that every vertex on the cycle has defect $2 - x$, and that the hub has defect $x|D|$. The result follows, since these two values are equal. $\square$

Proposition 4 *For $n \geq 3$, the minimum defect of the wheel is*

$$D(W_n, 2) = \frac{2\lceil n/3 \rceil}{\lceil n/3 \rceil + 1}.$$

Proof. We showed above that the state value is an upper bound. When $n = 3k$, the lower bound follows directly from Proposition 2. Indeed, $D(W_{3k}, 2) \geq D(F_{3k}, 2) = 2k/(k+1)$. So we need to establish the lower bound for $n = 3k+1$ and $n = 3k+2$.

Consider an optimal coloring of W_n with hub h and cycle $v_1, v_2, \dots, v_n, v_1$. By Lemma 2, we may assume there exist vertices u and u' such that $r(u) = 0$ and $r(u') = 1$. There are two cases.

(a) Assume $h \notin \{u, u'\}$. Then we can form $k-1$ edge-disjoint copies of F_3 avoiding u and u' . Let S denote the set of central vertices of the P_3 's of these copies. By Lemma 4, it follows that the total defect of $S \cup \{h\}$ within these copies is at least $2(k-1)$. Further, vertices u and u' together contribute defect 1 to the hub h . It follows that, in the graph as a whole, $d(h) + \sum_{s \in S} d(s) \geq 2k-1$, and so G has defect at least $(2k-1)/k$. If $k \geq 2$, then $(2k-1)/k \geq (2k+2)/(k+2)$, and we are done.

So consider the case when $k = 1$. Assume first that $n = 5$. Suppose u and u' are consecutive on the cycle; say $u = v_1$ and $u' = v_2$. Then $G - \{u, u'\}$ is a copy of F_3 . Since u and u' together contribute defect 1 to h , it follows from Lemma 4 that $d(h) + d(v_4) \geq 3$, and so G has defect at least $3/2$. So assume without loss of generality that $u = v_1$ and $u' = v_3$. If the hub has both two neighbors at least as red and two neighbors at most as red, then it has defect at least 2. So without loss of generality, we may assume that $r(v_2), r(v_4), r(v_5) \leq r(h)$. Then, the defect that h receives from $\{v_2, v_5\}$ is $2 - 2r(h) + r(v_2) + r(v_5)$, and the defect that u receives from $\{v_2, v_5\}$ is $2 - r(v_2) - r(v_5)$. That is, the sum of the defects that h and u receive from $\{v_2, v_5\}$ is at least 2. Since h also receives defect 1 from u and u' , it follows that $d(u) + d(h) \geq 3$, and the result follows. The argument for $n = 4$ is similar and omitted.

(b) Assume $h \in \{u, u'\}$. Say $u = v_1$ and $u' = h$. Consider v_n . It receives defect 1 from $\{v_1, h\}$ combined. Let index j be such that v_j is the redder vertex of v_{n-1} and v_n . Then v_{n-1} and v_n have at least $1 - r(v_j)$ of blue overlap and so $d(v_n) \geq 2 - r(v_j)$.

Further, one can choose k edge-disjoint copies of F_3 without using v_j . Let S denote the centers of the P_3 's of these copies. By Lemma 4 and noting that the hub h receives defect $r(v_j)$ from vertex v_j , it follows that $d(h) + \sum_{s \in S} d(s) \geq 2k + r(v_j)$.

Thus $d(h) + d(v_n) + \sum_{s \in S} d(s) \geq 2k + 2$, whence the result. $\square$

3.3 Complete multipartite graphs and expansions

We consider here complete multipartite graphs. These can be thought of as taking a complete graph and replacing each vertex by an independent set with the same adja-

gency. In general, we define $G[aK_1]$ to be the composition of G with the empty graph on a vertices; that is, the graph obtained by replacing every vertex v of G with a set I_v of size a such that a vertex of I_v is adjacent to a vertex of I_w if and only if v and w were adjacent. We let K_a^m denote the complete m -partite graph with a vertices in each partite set; that is $K_a^m = K_m[aK_1]$.

In some cases (for example, when the order of a complete graph is a multiple of k), the optimal coloring has every vertex with the same defect. In such a case, we can use the following simple idea:

Proposition 5 $TD(G[aK_1], k) \geq a^2 TD(G, k)$.

Proof. The bound follows by applying the lower bound to the a^n copies of G and averaging. $\square$

For example, it follows that:

Proposition 6 *If m is a multiple of k , then the complete multipartite graph K_a^m can be k -colored with defect $(m/k - 1)a$, and this is best possible.*

But if m is not a multiple of k , the result is not clear. Perhaps the following is true.

Conjecture 1 *The minimum defect in a k -coloring of K_a^m is $(\lceil m/k \rceil - 1)a$.*

In fact, we do not have an example that precludes it being the case that it always holds that $D(G[aK_1], k) = aD(G, k)$. Certainly this is an upper bound: take the optimal coloring of G and replicate it.

We now prove Conjecture 1 for 2 colors. We will need the following definitions. For a vertex x , define $L(x)$ as the number of neighbors such that $r(x) \geq r(y)$, and $S(x)$ as the number of neighbors such that $r(x) \leq r(y)$. Define a vertex x as **large** if $L(x) \geq S(x)$ and **small** if $L(x) \leq S(x)$. Note that

$$d(x) \geq \min(L(x), S(x)).$$

Proposition 7 *The complete multipartite graph K_a^m can be 2-colored with defect $(\lceil m/2 \rceil - 1)a$, and this is best possible.*

Proof. Such defect is attained by coloring all vertices in $\lfloor m/2 \rfloor$ of the partite sets with red, and the remaining vertices blue. So we need to prove that this is best possible.

If m is even, Proposition 5 shows that $TD(K_a^m) \geq m(m/2 - 1)a^2$, and thus some vertex has defect at least $(m/2 - 1)a$. So assume m is odd.

Assume first there is a partite set that contains both a large vertex u and a small vertex v . Note that we can increase $r(u)$ to 1 and decrease $r(v)$ to 0 without increasing the defect of either vertex. It follows that $d(u) + d(v)$ is at least their common degree, and thus $\max(d(u), d(v)) \geq (m - 1)a/2$.

So assume every partite set contains either only large vertices or only small vertices. Without loss of generality at least half the vertices are large. It follows that there are at least $(m + 1)a/2$ large vertices. Let u be the large vertex with *minimum* $r(u)$.

By the definition of large, $L(u) \geq (m - 1)a/2$. By the choice of u , at least $(m + 1)a/2 - 1$ vertices have red-component at least $r(u)$. Thus $S(u) \geq [(m + 1)a/2 - 1] - [a - 1] = (m - 1)a/2$. It follows that $d(u) \geq (m - 1)a/2$. $\square$

Corollary 1 $K_{a,a,c}$ with $a \leq c$ can be 2-colored with defect a , and this is best possible.

Proof. The minimum defect is attained by coloring all vertices in one of the smaller partite set with red, all vertices in another smaller partite set with blue, and all vertices in the largest partite set arbitrarily. The lower bound follows from Proposition 7, since $K(a, a, a)$ is a subgraph of $K(a, a, c)$. $\square$

We are unable to prove the following conjecture.

Conjecture 2 *The minimum defect of $K_{a,b,c}$ with $a \leq b \leq c$ for 2 colors is $bc/(b+c-a)$.*

This defect is attained by coloring all vertices v in the smallest partite set with $r(v) = 0$, all vertices in the largest partite set with $r(v) = 1$, and all vertices in the middle partite set with $r(v) = x$, where x is chosen to give the vertices in smallest and middle partite sets the same defect, namely $x = (b - a)/(b + c - a)$.

For another expansion, consider $C_m[aK_1]$ where m is odd. We now prove that $D(C_m[2K_1], 2) = 2$. What is perhaps surprising is that there are at least two different optimal colorings. The first such coloring is obtained by taking an optimal coloring

for C_m and replicating it. The second such coloring is obtained by, for each copy of $2K_1$, coloring one vertex red and one vertex blue.

Proposition 8 *For m odd, $D(C_m[2K_1], 2) = 2$.*

Proof. Consider a 2-coloring of $C_m[2K_1]$. We need to show that the defect is at least 2. As in the proof of Proposition 7, if a copy of $2K_1$ contains both a large vertex u and a small vertex v , then $d(u) + d(v) \geq 4$, so that the result follows. So we may assume that every copy of $2K_1$ contains either two large vertices or two small vertices. Since m is odd, it follows that there must be two adjacent copies of the same type. Assume u_1 and u_2 are adjacent to v_1 and v_2 with all four vertices large say. That is, $L(x) \geq 2$ for every $x \in \{u_1, u_2, v_1, v_2\}$. If any such x has $S(x) = 2$, the lower bound follows. Therefore we may assume that $S(x) \leq 1$ for every $x \in \{u_1, u_2, v_1, v_2\}$. This means that each u_i is redder than some v_j and vice versa, a contradiction. $\square$

3.4 Rooks graphs and Cartesian products

Recall that the Cartesian product of graphs G and H has vertex set $V(G) \times V(H)$ with vertex (g_1, h_1) adjacent to vertex (g_2, h_2) if and only if g_1 and g_2 are adjacent in G and h_1 and h_2 are adjacent in H .

We will need the obvious lower bound for the total defect of Cartesian products.

Proposition 9 *Let G and H be graphs of order m and n respectively. Then $TD(G \square H, k) \geq m TD(H, k) + n TD(G, k)$.*

Proof. The defect of a vertex in the product is the sum of the defects in its copies of G and H . $\square$

The **rooks graph**, denoted by $K_m \square K_n$, is the Cartesian product of the complete graphs K_m and K_n . We will denote the vertices by (i, j) with $1 \leq i \leq m, 1 \leq j \leq n$.

Lemma 5 *The rooks graph $K_m \square K_n$ can be 2-colored with defect $\lceil m/2 \rceil + \lceil n/2 \rceil - 2$.*

Proof. Color vertex (i, j) with red if i and j have the same parity and blue otherwise. $\square$

Corollary 2 *Let m and n be even integers. Then $D(K_m \square K_n, 2) = m/2 + n/2 - 2$.*

Proof. The upper bound follows from Lemma 5. The lower bound follows from Proposition 9, since $TD(K_s) = s(s/2 - 1)$ for s even, and thus $TD(K_m \square K_n, 2) \geq mn(n/2 - 1) + nm(m/2 - 1)$. $\square$

We show below that the upper bound in Lemma 5 is not always optimal. In fact we conjecture that it is never optimal when m and n are both odd, except for the case that $m = n = 3$.

Lemma 6 $D(K_3 \square K_3, 2) = 2$.

Proof. The upper bound is from Lemma 5.

We have two proofs of the lower bound, one by computer and one by hand. Both proofs entail converting the question to a set of linear programs. This is done by orienting each edge from smaller to larger proportion of red (with ties broken arbitrarily). If N_1 is the neighbors of vertex v with more red and N_2 is the neighbors of v with less red, then Equation 1 simplifies to $d(v) = |N_1|r(v) + |N_2|b(v) + \sum_{w \in N_2} r(w) + \sum_{w \in N_1} b(w)$, where $b(x) = 1 - r(x)$. Thus, minimizing the defect for a given orientation is a linear program. Further, if any vertex has in- and out-degree 2 for the orientation, the defect is definitely at least 2. With several pages of calculation or using a computer, one can show that $K_3 \square K_3$ has eight acyclic orientations (up to symmetry) that need to be considered, and then solve the eight associated linear programs. We omit the details. $\square$

In contrast, we found a coloring of $K_3 \square K_5$ that beats the bound of Lemma 5:

Lemma 7 $D(K_3 \square K_5, 2) \leq 38/13$.

Proof. A 2-coloring of $K_3 \square K_5$ is shown below. The element (i, j) of the matrix is the red-usage on vertex (i, j) .

$$\begin{bmatrix} 0 & 8/13 & 0 \\ 0 & 0 & 8/13 \\ 1 & 11/13 & 0 \\ 1 & 0 & 11/13 \\ 6/13 & 1 & 1 \end{bmatrix}$$

It can be verified that the defect of the coloring is $38/13$. $\square$

The above coloring can be extended to show that Lemma 5 is not optimal for $m = 3$ and n odd, $n \geq 5$, and indeed that $D(K_3 \square K_n, 2) \leq n/2 + 11/26$ in this case. However, this is still not best possible. For example, one can get defect $42/11$ for $K_3 \square K_7$ and defect $14/3$ for $K_3 \square K_9$ by the colorings illustrated:

$$\begin{bmatrix} 1 & 0 & 1 \\ 4/11 & 1 & 1 \\ 0 & 8/11 & 0 \\ 1 & 1 & 4/11 \\ 1/11 & 0 & 1 \\ 0 & 8/11 & 0 \\ 1 & 0 & 1/11 \end{bmatrix} \qquad \begin{bmatrix} 2/3 & 0 & 0 \\ 0 & 2/3 & 0 \\ 0 & 0 & 2/3 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

We used simulated annealing computer search to find upper bounds. Though we have no exact values, it seems to us that the heuristic computer search results suggest the following:

Conjecture 3 (a) If $m + n$ is odd, then $D(K_m \square K_n, 2) = (m + n - 3)/2$.
(b) If mn is odd and greater than 9, then $D(K_m \square K_n, 2) < \lceil m/2 \rceil + \lceil n/2 \rceil - 2$.

Note that Proposition 9 yields the following lower bounds:

Corollary 3

(a) If both m and n are odd, $D(K_m \square K_n, 2) \geq (m + n)/2 - 2 + 1/(2m) + 1/(2n)$.
(b) If m is even and n is odd, $D(K_m \square K_n, 2) \geq (m + n)/2 - 2 + 1/(2n)$.

For more colors we have one trivial observation: that $D(K_n \square G, k) = \lceil n/k \rceil - 1$ for any k -partite graph G , as a corollary of Proposition 1.

3.5 Regular graphs

Lovász [6] showed that we can ordinary 2-color a cubic graph with defect at most 1. Therefore $D(G, 2) = 1$ for all nonbipartite cubic graphs G .

For a 4-regular graph, Lovász's result shows that one can ordinarily 2-color it with defect at most 2. We conjecture that this can be improved. Proposition 8 shows that the expansion $G = C_m[2K_1]$ where m is odd has $D(G, 2) = 2$. Using simulated annealing (that is, a randomized search for a coloring), the computer can find a 2-coloring with defect smaller than 2 for all 4-regular graphs on up to 14 vertices, except for the expansions of odd cycles, and the two graphs K_5 and $K_3 \square K_3$, which we saw earlier have minimum defect 2. We conjecture a general behavior:

Conjecture 4 *Apart from $G = C_m[2K_1]$ where m is odd, it holds that $D(G) < 2$ for all but finitely many connected 4-regular graphs.*

4 Complexity

Unsurprisingly, it is NP-hard to determine if there is a coloring with defect at most some specified d .

One way to see this is that fractional defect 2-coloring is NP-hard even for $d = 1$. One can extend Lemma 2 to show that in graphs of minimum degree at least 3, a 2-coloring with defect 1 can only be a coloring with monochromatic vertices. Thus the fractional defect 2-coloring problem is equivalent to the ordinary defective 2-coloring problem in such graphs. The latter problem was shown to be NP-hard by Cowen [3]. (Actually, we need ordinary 1-defect coloring to be NP-hard in graphs with minimum degree at least 3. But one can transform a graph to having minimum degree at least 3 without changing the coloring property by adding, for each vertex v , a copy of K_4 and joining v to one vertex of the K_4 .)

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